# Exciton torque

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Useful references and info about the exciton in tj model

## I. USEFUL REFERENCES

[1] They compute exciton spectra and optical conductivity using t-J model on a four-leg cylinder using MPS (Their model is the typical t-j model with Hubbard and J between nearest neighbors), starting from half-filling ground state and applying an exciton operator, where, depending on the dopant (fermionic, bosonic, or a mixture) the corresponding excitonic operator is defined. More details about the MPS simulations are in Appendix C. Their ultimate goal is to compare against the Fermi-Hubbard and understand the nature of their excitations, e.g., exciton vs distinguishable dopants, allowing the tracking of the nature of the Fermi-Hubbard excitons.

## II. MODEL AND METHOD

We consider an effective electron-hole model with spin degrees of freedom coupled to localized semiclassical spins

$$\hat{H} = \hat{H}_0 + \hat{H}_{sd}[\mathbf{M}(t)] + \hat{H}_U + \hat{H}_V \tag{1}$$

$$\hat{H}_0 = \sum_{\langle ij\rangle,l} \hat{c}_i^{l\dagger} H^l_{0,ij} \hat{c}^l_j \tag{2}$$

$$\hat{H}_U = U \sum_{il} \hat{n}^l_{i\uparrow}\hat{n}^l_{i\downarrow} \tag{3}$$

$$\hat{H}_V = -V \sum_{\sigma\sigma' l} \hat{n}^l_{i\sigma}\hat{n}^l_{i\sigma'} \tag{4}$$

$$\hat{H}_{sd}(t) = -J_{sd} \sum_{i,l} \mathbf{M}^l_i(t)\cdot \hat{\mathbf{s}}^l_i \tag{5}$$

The standard semiclassical dynamics for the LMMs coupled to electrons is obtained as a saddle point of the Gaussian approximation of the effective action (once the electrons are integrated out), which only works at small coupling $j_{sd} \ll \gamma$, obtaining an effective nonlocal damping. In this approximation, the Kernel is obtained from the magnetic susceptibility of the electrons:

$$\begin{aligned} R^{ab}_{ij}(t) &= J^2_{sd}\int_{-\infty}^{t} dt' \chi^{ab,R}_{ij}(t-t') && (6)\\ \chi^{ab,R}_{ij}(t,t') &= -i\theta(t-t')\langle[\hat{s}^a_i(t), \hat{s}^b_i(t')]\rangle && (7)\end{aligned}$$

Thus by exact methods, it is possible to get full suceptibility. On the other hand, once the excitonic effects are included...

The bare dynamics of $\mathbf{M}^l_i(t)$ is described by the classical hamiltonian $H_{cl}[\mathbf{M}]$. Our goal is to derive an effective semiclassical dynamics for the LMMs, including the effects of correlated electrons within each layer and exciton formation in between layers.

$$\partial_t \mathbf{M}^l(t) = \mathbf{M}^l(t)\times\Big[\mathbf{B}^l_{cl}[\{\mathbf{M}(t)\}] + \mathbf{B}^l_{\Phi}[\{\mathbf{M}(t)\}] + \mathbf{B}^l_0[\{\mathbf{M}(t)\}]\Big] \tag{8}$$

$$\mathbf{B}^{l,cl}[\mathbf{M}(t)] = -\frac{\partial H^{cl}[\{\mathbf{M}(t)\}]}{\partial \mathbf{M}^l(t)} \tag{9}$$

### A. Excitonic torque ($U = 0$)

We write the effective action as a function of the intralayer excitonic order parameter $\hat{B}^{\sigma\sigma'}_i = \hat{c}^{1\dagger}_{i\sigma}\hat{c}^2_{i\sigma'}$, using $\hat{n}^1_i\hat{n}^2_i = 2\hat{n}^1_i - \sum_{\sigma,\sigma'}\hat{B}^{\sigma\sigma'}_i\hat{B}^{\sigma\sigma'}_i$ and absorbing the first term into the chemical potential, the electronic action $\mathcal{S}_e$ reads

$$\mathcal{S}_e = \int_{\mathcal{C}} dz dz' \sum_{ij} \bar{\Psi}^T_i G^{-1}_{\mathbf{M}} \Psi_j + V\int_{\mathcal{C}} dz \sum_{i\nu} B^\nu_i \bar{B}^\nu_i, \tag{10}$$

where $\Psi = (c^1_\uparrow, c^1_\downarrow, c^2_\uparrow, c^2_\downarrow)^T$, and the polarized Green function (GF) satisfies $\big(i\partial_z - H_0 - H_{sd}[\mathbf{M}]\big)G_{\mathbf{M}}(z,z') = \delta_C(z,z')$. We additionally decompose the excitonic field on its charge and spin degrees of freedom using $B^\nu_i = \mathrm{Tr}_\sigma[\Gamma^\nu B_i]$, with $\Gamma^\nu = \sigma^\nu/\sqrt{2}$ and $\nu = 0,x,y,z$. Applying the Hubbard Strattonovich (HS) transformation $\mathcal{S}_V \to \mathcal{S}^{HS}_V[\{\Phi^\nu_i\}]$, the action is written as

$$\begin{aligned} \mathcal{S}_e &= \int_{\mathcal{C}} dz dz' \sum_{ij} \bar{\Psi}^T_i (G^{-1}_{\mathbf{M},ij} - \delta_{ij}\mathcal{V}_\Phi)_{ij}\Psi_j \\ &+ \frac{1}{V}\int_{\mathcal{C}} dz \sum_{i,\nu} \bar{\Phi}^\nu_{\mathrm{i}}\Phi^\nu_{\mathrm{i}}. && (11)\end{aligned}$$

Integrating out the electrons and keeping to second order (RPA), the effective exciton action reads.

$$\mathcal{S}^{\text{eff}}_{\Phi}[\mathbf{M}] = \int_{\mathcal{C}} dzdz' \sum_{\nu\nu'} \bar{\Phi}^{\nu} \mathrm{D}^{-1}_{\nu\nu'} \Phi^{\nu'} - \mathrm{iTrln}[\mathrm{G}^{-1}_{\mathbf{M}}] \quad (12)$$

where we assumed a homogeneous order parameter for the exciton, the propagator of the exciton satisfies the Bethe-Salpeter equation $D = V(1 + \Pi D)$. Note that linear term gave us zero because there is not a direct mechanism to generate the exciton in the hamiltonian $t_{\perp}$

$$D^{-1}_{\mu\nu}(z, z') = \frac{\delta_{\mathcal{C}}(z, z')\delta_{\mu\nu}}{V} - \Pi_{\mu\nu}(z, z') \quad (13)$$

$$\Pi_{\mu\nu}(z, z') = -\frac{i}{L}\mathrm{Tr}[\Gamma^{\mu\dagger} G^1_{\mathbf{M}}(z, z')\Gamma^{\nu} G^2_{\mathbf{M}}(z', z)], \quad (14)$$

Additionally, we integrate out the exciton degrees of freedom to get an effective action for the classical moments. The total effective action reads,

$$\mathcal{S}^{\text{eff}}_{\text{cl}} = \mathcal{S}_{cl} - i\mathrm{Trln}[G^{-1}_{\mathbf{M}}] + i\mathrm{Trln}[D^{-1}_{\mathbf{M}}] \quad (15)$$

To find the semiclassical dynamics, we minimize with respect to the classical LLMs $\frac{\partial \mathcal{S}^{\text{eff}}_{\text{cl}}}{\partial \mathbf{M}^l} = 0$, the classical term, which contains the Berry phase and its respective classical Hamiltonian of the LMMs returns

$$\frac{\partial \mathcal{S}_{cl}}{\partial \mathbf{M}^l(t)} = \frac{1}{\gamma_l}\mathbf{M}^l \times \partial_t \mathbf{M}^l + \mathbf{B}^{l,cl} \quad (16)$$

The contributions to the effective field from the polarized electrons is

$$\mathbf{B}^l_0[\{\mathbf{M}(t)\}] = i\frac{\partial \mathrm{Trln}[G^{-1}_{\mathbf{M}}]}{\partial \mathbf{M}^l} = J_{sd}\langle \hat{\mathbf{s}}^l \rangle_{\mathbf{M}} \quad (17)$$

and the contribution from the exciton is

$$\mathbf{B}^l_{\Phi}[\{\mathbf{M}(t)\} = -i\frac{\partial \mathrm{Trln}[D^{-1}]}{\partial \mathbf{M}^l} = -i\mathrm{Tr}\Big[D\frac{\partial \Pi}{\partial \mathbf{M}^l}\Big] \quad (18)$$

Where the explicit form of the torques is: (To compute the real-time dynamics, we should apply Langreth rules to get the retarded component):

$$B^1_{\alpha\Phi}(\bar{z}) = \int dzdz' \frac{j_{sd}}{L}\sum_{\mu\nu} D_{\mu\nu}(z, z')\mathrm{Tr}\big[\Gamma^{\mu} G^2_{\mathbf{M}}(z', \bar{z})\Gamma^{\nu} G^1_{\mathbf{M}}(\bar{z}, \bar{z}')\sigma^{\alpha} G^1_{\mathbf{M}}(z', z)\big] \quad (19)$$

$$B^2_{\alpha\Phi}(\bar{z}) = \int dzdz' \frac{j_{sd}}{L}\sum_{\mu\nu} D_{\mu\nu}(z, z')\mathrm{Tr}\big[\Gamma^{\mu} G^2_{\mathbf{M}}(z', \bar{z})\sigma^{\alpha} G^2_{\mathbf{M}}(\bar{z}, z)\Gamma^{\nu} G^1_{\mathbf{M}}(z, z')\big]$$

Putting all together, and following the standard procedures, we finally arrive to Eq. [(**??**)]

Up to this point, the theory is completely self-consistent, $\mathbf{M} \to G_{\mathbf{M}} \to D \to \mathbf{M}$. At each time step, the classical magnetization, the polarized GF, and the exciton propagator should be evolved self-consistently. There are two paths that can be followed:

- Given a classical trajectory of the LMMs, calculate the exciton response and spectra from $D[\mathbf{M}]$
- Use a $j_{sd}$ expansion to get explicit expressions for torques without self-consistency

### 1. $j_{sd}$ *expansion and LLG equations*

In order to get the standard LLG, the effective fields should be expanded to first order in M. This can be done in two different ways:

- Considering and adiabatic approach, where $G_{\mathbf{M}}$ is computed explicitly from a frozen configuration. (It sounds like Born-Oppenheimer nucleus-electron motion), and still evolve self-consistently
- Expanding to first order $G_{\mathbf{M}} = j_{sd}\mathbf{M}G^{(1)} + O(j^2_{sd})$ and eliminate the selfconsistency because the kernels can be computes independently

The second way emerges naturally as a second-order expansion of $S^{\text{eff}}$, and is the natural way to get the effective LLG equation type. In our case, the susceptibility comes from

$$(j_{sd})^2\chi^{ll'}_{ab}(z, z') = \frac{\partial \mathcal{S}^{\text{eff}}}{\partial M^l_a(z)\partial M^{l'}_b(z')} \quad (20)$$

For the electronic mediated kernel, the standard nonlocal kernel, where only the retarded component contributes to the equations

$$R^{0,ll'}_{ab}(z, z') = (j_{sd})^2\delta_{ll'}\sum_k \mathrm{Tr}[\sigma^a g^l_k(z, z')\sigma^b g^l_k(z', z)] \quad (21)$$

$$= (j_{sd})^2\delta_{ll'}\delta_{ab}\sum_k g^l_k(z, z')g^l_k(z', z) \quad (22)$$

The susceptibility contribution from the exciton is:

$$R_{ab}^{\Phi(I),ll'}(z,z') = -i\mathrm{Tr}_\nu[D\circ\Lambda_{ab}^{ll'}(z,z')] \quad (23)$$
$$R_{ab}^{\Phi(II),ll'}(z,z') = i\mathrm{Tr}_\nu[D\circ\Lambda_l^a(z)\circ D\circ\Lambda_{l'}^b(z')] \quad (24)$$

where the vertex functions $\Lambda_a^l(z,z';\bar{z}) = \frac{\partial\Pi_\mu(z,z')}{\partial M_a^l(\bar{z})}$ are:

$$\Lambda_a^1(z,z';\bar{z}) = i\frac{J_{sd}}{L}\sum_k \mathrm{Tr}\big[\Gamma^\mu G_{\mathbf{M}}^2(z,z')\Gamma^\nu G_{\mathbf{M}}^1(z',\bar{z})\sigma^a G_{\mathbf{M}}^1(\bar{z},z)\big]$$

$$\Lambda_a^2(z,z';\bar{z}) = i\frac{J_{sd}}{L}\sum_k \mathrm{Tr}\big[\Gamma^\mu G_{\mathbf{M}}^2(z,z')\sigma^a G_{\mathbf{M}}^2(z',\bar{z})\Gamma^\nu G_{\mathbf{M}}^1(\bar{z},z)\big]$$

$\Lambda_a^l(z')$ denotes the 1 vertexes fot the 2 vertexes we got

$$\Lambda_{\mu\nu;ab}^{21}(z,z';\bar{z},\bar{z}') = -i\frac{j_{sd}^2}{L}\mathrm{Tr}\Big[\Gamma^\mu G_{\mathbf{M}}^2(z,\bar{z})\sigma_a G_{\mathbf{M}}^2(\bar{z},z')\Gamma^\nu G_{\mathbf{M}}^1(z',\bar{z}')\sigma_b G_{\mathbf{M}}^1(\bar{z}',z)\Big] \quad (25)$$

Studying the polarized exciton propagator $D_\Phi(\omega;\{\mathbf{M}\})$, despite the classical and exciton degrees of freedom had a completely different energy scales $\omega_{\mathrm{mag}} \ll \omega_\Phi$, we evidence the emergence of the bound exciton at different excited energies and its dependence with the adiabatic magnetic configurations of the classical LMMs. The exciton behavior is strongly dependent on the adiabatic magnetic configuration, generating a mechanism that connects both degrees of freedom. E.g., exciting the exciton with light (or other medium) will be reflected in the effective coupling of the LMMs in between layers $J_{\mathrm{eff}}^\Phi\mathbf{S}_1\cdot\mathbf{S}_2$ Note also that there is no need for coherent exciton resonance $\langle B\rangle = 0$ to feel the exciton effects, which is similar to thinking about it as a noise(generating damping, memory, and retardation effects). But it is also possible to generate an explicit coherent exciton $\langle B\rangle \neq 0$ which can condensate dynamically (Millis) by light pulse or an injection of excitons. Without expanding, all this info should be contained into D allowing an experimental platform to track optical excitations with magnetic degrees of freedom (Bae and so... have used this mechanism to read optical excitations from magnetic probes).

A natural question emerges about the opposite mechanism. Is it possible to generate exciton signatures through classical magnetic excitations? Probably the answer is yes?? This excitation depends on two possible mechanisms: not sure... but shouls be something similar to [2, 3] Effective equation for the exciton which condensates it??

- Incoherent response of the excitons under adiabatic variations of exited modes of LMMs $\mathbf{M}$. $\delta\Phi(\omega_{ext}) = \delta\Phi_{ad}(\omega_{ext})$
- Injection of excitons, which is equivalent to having an effective displacement of the chemical potential in between layers $\mu_\Phi$. This will allow us to reduce the exciton resonance to a lower frequencies $\omega_\Phi \approx \Delta - \mu_\Phi \approx \omega_{\mathrm{mag}}$ and excite by external sources a magnetic mode of the classical LMMs

Thus, without need of an optical drive (only magnetic and electronic mediums are needed to excite and generate a real crossover of both degrees of freedom) a coherent excitation will emerge in between both.

Is important to clarify the difference between both mechanism. In the first one, the exciton order parameter is generated by an optical excitation, but there is not real interplay between exciton and magnetic degrees of freedom, instead the magnetization just "feel" the exciton. Instead, in the second one, the external drive comes from a magnetic excitation Maybe the general way to include both excitations in general is including a drive in the action $F(\omega) \propto \delta(\omega - \omega_0^{\mathrm{crsbr}})$? or just assuming the classical LMMs are oscillating at that given frequency..?? not sure.... If we include the driving and solve for $M(\omega\to\omega_0)$ and later use that $M$ to solve for $\Phi$, is this correct?

Thus, to see the exciton response, we should find $\frac{\delta S}{\delta\Phi} = 0$, which through $\mathbf{M}$ in $D$ gave us an effective excitation mechanism to generate exciton (in the action should appear a term as $F\Phi + h.c$), lets derive this...

The excitonic resonance thus corresponds to

$$\int D^{-1}(t,t')\Phi(t') = 0 \quad (26)$$

which has nonzero solutions only if $\det[D^{-1}(\omega')] = 0$. We can use the minimum energy solution and expand around it $\Phi = \Phi_{\mathrm{adi}} + \delta\Phi$ (Note that we can justify it saying that we are expanding around the adiabatic background, this does not necesarily satisfies always the condition, thus linear terms are not necessarily zero). The effective action for the excitons thus reads:

$$\delta\bar{\Phi}\big[D^{-1}\big]\delta\Phi + \delta\Phi\big[D^{-1}\Phi_0\big] + \mathrm{h.c} \quad (27)$$

Thus, the adiabatic background acts as an effective source of the fluctuations of the excitonic order parameter with

$$\left[D^{-1}\Phi_0\right] = \left(D^{-1}_{\mathbf{M}_0}\Phi_0 = 0\right) + D^{-1}_{\delta\mathbf{M}}\Phi_0 \tag{28}$$

minimizing the fluctuations, we see that adiabatic corrections on a static background act as a source term for the exciton (compare with the dipolar term that emerges from ligth )

$$\begin{aligned} \left[D^{-1}\right]\delta\Phi &= -D^{-1}_{\delta\mathbf{M}}\Phi_0 & (29)\\ \delta\Phi &= -DD^{-1}_{\delta\mathbf{M}}\Phi_0 & (30)\\ &= (DD^{-1}_{\mathbf{M}_0} - 1)\Phi_0 & (31) \end{aligned}$$

Note that if the full response is adibatic $DD^{-1}_{\delta\mathbf{M}} = 1$ the corrections are zero

once we get the adiabatic response $\Phi = \Phi_{ad} + \delta\Phi$ way to think responses is in terms of variations $\delta\Phi = \delta\Phi_{ad} + \delta(\delta\Phi)$? being this the case thus maybe the first term is enogh to consider the effects of the magnetic excitation on the excitons?

In some way this are corrections of the adiabatic response and are self induced effective fields, im not completely sure how to deal with them. to warm up, lets instead to introduce and external drive that couples to the exciton degree of freedom and study their effects

## III. LIGHT INDUCED EXCITON TORQUE

Lets add light that couple to the exciton degree of freedom through a dipolar term

$$\mathcal{S}^{\Phi}_{\text{ligth}} = -\int dz \sum_{\mu}(\bar{\Phi}^{\mu}F^{\mu} + \bar{F}^{\mu}\Phi^{mu}) \tag{32}$$

For simplicity we assume non polarized light $F^{\mu} = (F(t), 0, 0, 0)$, where $F(t) = Ae^{-(t-t_0)^2/(2\sigma)^2}\sin(\omega t)$ is a gaussian pulse. Looking to the saddle point, we obtain an equation for $\Phi^0$

$$\Phi^0(z) = \int dz' D_{\mathbf{M}}(z, z')F(z') \tag{33}$$

Considering only the zeroth order term the effective action reads:

$$\mathcal{S}^{\text{eff}}_{\text{cl}} = \mathcal{S}_{cl} - i\text{Trln}[G^{-1}_{\mathbf{M}}] - \text{Tr}\left[F\mathcal{R}_{\mathbf{M}}F\right] \tag{34}$$

where $\mathcal{R}_{\mathbf{M}} = (D_{\mathbf{M}} + D^{\dagger}_{\mathbf{M}})/2$ is the real part of the exciton propagator. We calculate the polarized susceptibility from the new exciton term. Note that the contribution looks different than before and came uniquely from the coherent part of the exciton $\Phi = \langle\hat{B}\rangle$

$$R^{I}_{12}(z, z') = \text{Tr}\left[F\mathcal{R} \circ \Lambda^{I}_{1}(z) \circ \mathcal{R} \circ \Lambda^{I}_{2}(z') \circ \mathcal{R}F\right] + \leftrightarrow 21 \tag{35}$$

$$R^{II}_{12}(z, z') = \text{Tr}\left[F\mathcal{R} \circ \Lambda^{II}_{12}(z, z') \circ \mathcal{R}F\right] \tag{36}$$


[1] A. Bohrdt, E. Demler, and F. Grusdt, Spectroscopy of hubbard-mott excitons and their ro-vibrational excitations, arXiv preprint arXiv:2406.16854 (2024).

[2] E. Perfetto, K. Wu, and G. Stefanucci, Theory of coherent phonons coupled to excitons, npj 2D Materials and Applications **8**, 40 (2024).

[3] D. Golež, Z. Sun, Y. Murakami, A. Georges, and A. J. Millis, Nonlinear spectroscopy of collective modes in an excitonic insulator, Physical Review Letters **125**, 257601 (2020).